

ORIGINAL RESEARCH

Trustworthiness and transparency features were less frequent in randomized trials presenting large effects for continuous outcomes in abstracts

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Abstract

Objectives: Large effect sizes (ESs), especially when prominently presented in trial abstracts, draw large attention, but it is important to understand whether they are trustworthy. We aimed to assess indicators of transparency and trustworthiness in randomized controlled trials (RCTs) reporting some large ES for continuous outcomes in their abstract, in comparison with RCTs presenting only nonlarge ESs in their abstract.

Study Design and Setting: We included RCTs indexed in MEDLINE between January 1, 2024 and March 18, 2025, presenting at least one standardized mean differences of absolute value 0.8 or higher (large ES) vs those presenting only smaller absolute standardized mean differences in their abstract. Trial characteristics and methodological features were extracted systematically in large ES and nonlarge ES trials. Primary outcome was prespecified protocol registration, secondary outcomes were any protocol registration and public availability or repository placement of raw data (preregistered protocol: <https://osf.io/8xasw>).

Results: We evaluated 152 trials with large ESs in their abstract and 175 trials with only nonlarge ESs in their abstract. Large ES trials had suggestively lower rates of preregistered protocols (45% vs 61%, $P = .0054$) and significantly lower rates of any protocol registration (74% vs 87%, $P = .0028$) than nonlarge ES trials. There was no difference in raw data public availability or repository placement (6% vs 7%). Large ES trials were also less likely to be multicenter ($P = .0042$), to have high-income country of corresponding author ($P = .0001$), to be conducted in high-income country site(s) ($P = .0003$), to have a published statistical analysis plan ($P = .0216$), to result from between-group comparisons ($P < .0001$), and tended to report less frequently allocation concealment ($P = .0351$). Large effects were significantly more likely to involve nondrug/nonpsychological interventions ($P = .0001$).

Conclusion: RCTs presenting large ESs in their abstracts are more likely to lack transparency and trustworthiness features and may operate with higher risk of lack of credibility. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Meta-research; Epidemiology; Transparency; Methodology; Data sharing; Preregistration; Reproducibility

1. Introduction

Large effect sizes (ESs) in clinical trials draw attention and may be desirable for investigators and journals to publish. However, an important question is whether they can be trusted. A large ES may be too large to be true, that

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Plain Language Summary

When a medical study reports that a treatment has a big effect that naturally grabs attention. However, are impressive-sounding results actually reliable? This study investigated whether clinical trials claiming large treatment effects in their abstracts are as trustworthy as those reporting only more modest results. We compared two groups of recently published randomized controlled trials: 152 trials that reported large effects in their abstracts, and 175 trials that reported only smaller effects. We looked at several markers of good scientific practice, particularly whether researchers had registered their study plans in advance and whether they made their data openly available. We found that trials reporting large effects were less likely to have preregistered their methods and more likely to have no registered protocol at all. The two groups also differed in other aspects of trustworthiness. Large-effect trials were less likely to involve multiple research centers, less likely to have researchers from high-income countries, and less likely to have a published statistical analysis plan. They also tended to test different types of treatments—large effects appeared more often in studies of interventions other than drugs or psychological therapies. Both groups were equally poor at making their raw data publicly available (only about 6%-7%). Overall, when a trial claims to have found impressive treatment effects, some extra scrutiny may be warranted. Before trusting eye-catching results, readers should assess whether trials followed proper scientific checks and transparency safeguards.

is, it may be spurious, inflated, and occasionally even fraudulent or fake [1–3]. Previous work suggests that in some fields large ES claims are rarely replicated [4]. In an evaluation of over 85,000 Cochrane meta-analyses, effect sizes exceeding 5 on the odds ratio (OR) scale were largely observed in very small trials and they almost always diminished or completely evaporated when tested in additional larger randomized controlled trials (RCTs) [5]. It is also established that abstracts tend to report extreme results [6], as investigators may hype the most prominent findings [7,8]. Nonsignificant results may be presented as significant in the abstract [9] and larger ESs may also be selected for showcasing in the abstract. If so, focusing on abstracts might give an enriched sample in claims of large ESs. Abstracts are the most read part of any publication, including RCTs, and they are most influential in shaping perceptions about the results and importance of the presented work. Favorable spin can further shape clinicians' interpretation of the evidence and thus decision-making [10].

Protocol preregistration and demonstrable availability of raw data may offer strong support for the trustworthiness and transparency of an RCT [11,12]. Here, we aimed to assess empirically whether protocol registration (in particular prospective registration) and data availability are largely absent in trials that prominently report large ESs in their abstracts. For contrast, we assessed how these trials differ vs RCTs that also report numerically only more modest ES(s) in their abstract.

Given that transparency practices of preregistration and data availability have only recently become modestly common [13], we focused on very recent RCTs. Assessment of the more recent practice may then also offer evidence that is relevant for the current state of RCTs and their potential trustworthiness. We further aimed to explore additional design or other features that also differentiate trials with and without large reported ESs.

2. Methods

2.1. Protocol

This is a meta-epidemiological assessment with a case-control design of RCTs that present ESs in the form of standardized mean differences (SMDs) in their abstract. The protocol of this study has been preregistered and published at the Open Science Framework (<https://osf.io/8xasw>). Methods followed the Preferred Reporting Items for Systematic reviews and Meta-Analyses reporting guidelines, including the adaptations for meta-epidemiological studies and checklist [14]. The methods are described in further detail in the published protocol. Clarifications to the prespecified methodology were required during the conduct of the analysis in some cases, as described below.

2.2. Eligible studies

We searched the PubMed database, and selected RCTs published between January 1, 2024 and March 18, 2025, that reported at least one large ES in their abstracts; and a control group of RCTs reporting only nonlarge ESs in their abstracts. The exact entry terms for the database search are in the published protocol: <https://osf.io/8xasw>. We only included RCTs reporting SMD or related measures, to be consistent with the choice of effect metrics. ESs with a magnitude of SMD, Cohen's *d* or Hedges' *g* of 0.8 or higher in absolute terms (positive or negative) were regarded as large. For each eligible trial abstract, we selected the largest reported ES in absolute magnitude to represent it.

2.3. Data extraction

We extracted prespecified key methodological characteristics. We assessed a pilot sample of data (40 with large ES and 40 with nonlarge ES) to inform the construction of the data extraction form and sample size calculations. After

What is new?**Key findings**

- This meta-research study indicated that RCTs presenting large effect sizes in their abstracts are more likely to lack features of transparency and trustworthiness.

What this adds to what was known?

- Large effect sizes in abstracts of trials draw attention and may seem as compelling evidence for a study's claim. Yet there have been divergent findings concerning the association between effect size and credibility of a trial. Our study showed that large effect sizes are consistently associated with a range of methodological shortcomings.

What is the implication and what should change now?

- RCTs presenting large effect sizes may operate with higher risk of lack of credibility.
- Rather than indicating a convincing finding, extreme results should be met with increased attention and careful scrutiny.

discussion, a final list of data items for extraction and analysis was defined. To secure interinvestigator reliability, the data extraction process was independently replicated by additional reviewers for an initial pilot subset of 80 trials (J.R.P. and L.K.). Any discrepancies between reviewers were resolved through discussion among the research team and a refined data extraction protocol was established. The resulting data extraction protocol was then subsequently applied to the complete dataset of included trials by one reviewer (J.H.). To verify the accuracy of our manual data extraction, we conducted automated text mining using the R packages *oddpub* and *rtransparent*. PDF files of included studies were first converted to text files using *oddpub* and then automatically screened for statements on data availability (*oddpub*) as well as study registration (*rtransparent*). The automated results were then cross-validated against our manually extracted data.

2.4. Outcomes and statistical analysis

Primary outcome measure was the prospective protocol registration in large ES and nonlarge ES groups ("preregistration"), defined as registration within 14 days of onset of study recruitment, as suggested previously [3]. With no given study conduct/recruitment dates and/or no given registration date, studies were classified as not-preregistered. Whenever only month of study initiation or recruitment was given, protocol registration was defined

as preregistered only if registration was no later than within that same month.

Some issues arose during data collection requiring clarifications to our original protocol. We initially collected information of start dates from the published manuscript and extracted information on the protocol registration date as defined in the respective protocol registry database. Any information provided in the original article concerning protocol registration was used to search for and identify the corresponding protocol, that is, protocol registry, registry number, and/or URL. However, a study initiation or recruitment date was often not stated in the published manuscript, so we also checked for this information in the trial registries. The presented analyses include all available information from these sources. When there was a discrepancy between the manuscript and registry information on whether the trial was prospectively registered, we preferred the manuscript information. We also performed a sensitivity analysis where we preferred the registry-listed information and another sensitivity analysis where we considered all trials with contradictory information to be retrospectively registered. Statistical significance level for the primary outcome was set at $\alpha = 0.005$, as per ref. [15] where *P* values between 0.05 and 0.005 are considered suggestive.

Secondary outcome measures were any protocol registration (prospective or retrospective) and availability of (deidentified) individual participant data (categorized as already publicly accessible now or already posted in a repository). We extracted information on data availability as stated anywhere in the published trial manuscript. When public availability or repository placement of data was claimed in the publication, this was checked and verified for each individual case. To be categorized as "repository placement of data", the manuscript had to provide information on the data repository and, when data were not accessible (eg, without further request), the data repository had to contain at least the information that data of the same exact trial was stored. *P* values threshold for secondary outcomes was set at 0.05.

For all categorical outcomes, we derived risk ratios and 95% confidence intervals (CIs) and *P* values using Fisher's exact tests. For continuous outcomes, we derived median and interquartile ranges (IQRs) and *P* values using Mann-Whitney U tests.

Additional analyses explored (using a *P* value threshold of 0.05 without multiplicity adjustment) other features that may also reflect trustworthiness and transparency: availability of statistical analysis plan (SAP), corresponding author with any retracted article(s) (identified through Retraction Watch [16]; for authors with common names, publications were cross-referenced between Scopus and Retraction Watch databases to ensure accurate attribution to the specific author [17], any data sharing statement in article, statement of data availability upon request, and statement that data will not be shared. We searched for information on SAP within the trial publication and the protocol registry. The SAP had to be a separate document from

the paper and/or a separate explicitly titled SAP after an overall protocol. Having simply a section of “statistical analysis” in the paper or protocol would not qualify. We also examined if groups differed in the following characteristics: sample size, ES referring to between-group comparison of the randomized arms (as opposed to within-group comparison in one of the arms or isolated indeterminate ES, a subset or the total study population), statistical significance of the defining largest ES (ie, declaration of $P < .05$, or verbal quote of significance, or 95% CI excluding the null, in the abstract, referring to the largest ES that represents the RCT), corresponding author from high-income country [18], study conducted in high-income country, multicenter trial, journal impact factor, research intervention category (drug/biologic/vaccine, psychological/behavioral/educational, other (eg, invasive procedures and physical exercise interventions)). Following suggestions by a peer-reviewer, we also examined presence of blinding, allocation concealment, objective outcome, and early stopping due to effectiveness in interim analyses.

In subgroup analyses, primary and secondary outcome analyses were stratified by country, multicentricity,

research intervention, and journal impact factor (above/below median). Differences in between subgroup ORs of primary and secondary outcomes were assessed using Cochrane’s Q test. Multivariable logistic regression analysis considered all factors with $P < .05$ in univariable associations, in addition to the large/nonlarge ES group membership in assessing protocol preregistration and no protocol availability as dependent variables. Additional sensitivity analyses of primary and secondary outcomes considered alternative ES thresholds, that is, SMD of 0.5, 1.0 and 1.5.

All analyses were conducted using R and R Studio, Version 4.5.1, Posit Software, PBC.

3. Results

3.1. Eligible RCTs assessed

The MEDLINE database search retrieved a final set of 327 trials that reported a numerical effect size in their abstract—152 RCTs presenting a large ES (0.8 or higher), and 175 RCTs presenting only nonlarge ES (Fig Flow diagram).

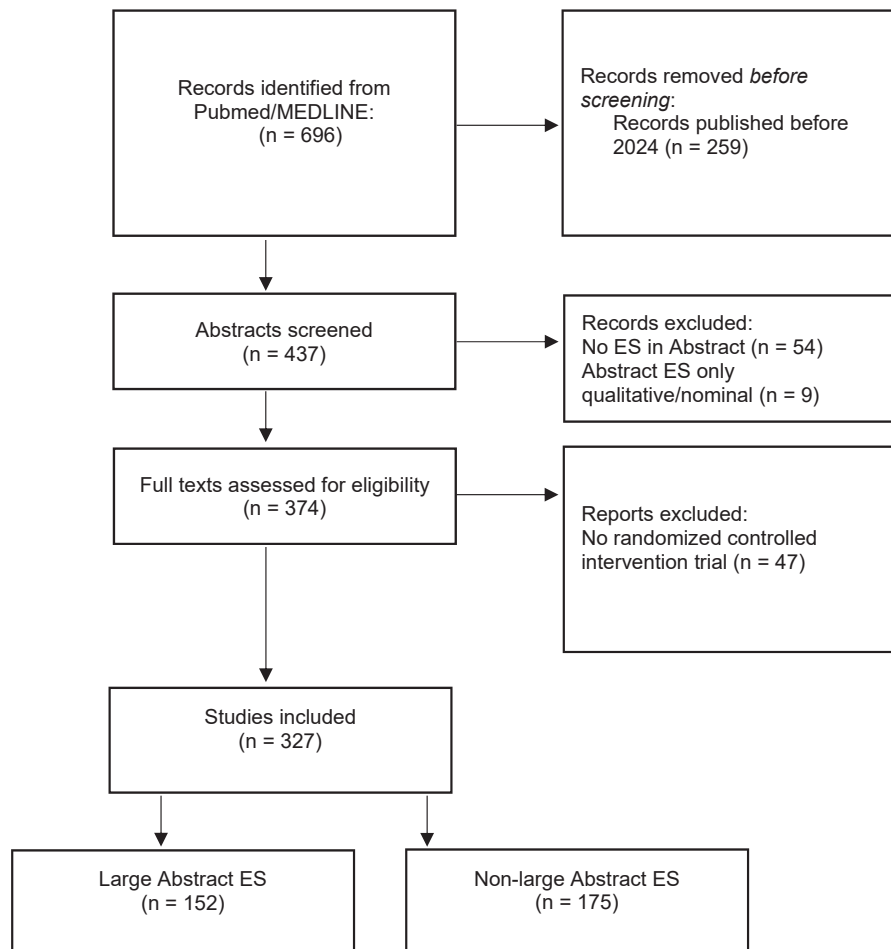


Figure. Flow diagram: Identification of records and inclusion of studies—adapted from PRISMA 2020 [19]. ES, effect size.

All 327 included trials had a median sample size of 92 (IQR: 43.5–191) and a median absolute ES of 0.7 (IQR: 0.5–1.2), when selecting the largest absolute ES reported. Large ES trials had a median sample size of 60 (IQR: 32–110) and a median absolute ES of 1.2 (IQR: 1–1.9). Nonlarge ES trials had a median sample size of 137 (IQR: 71–262) and a median absolute ES of 0.5 (IQR: 0.4–0.6).

3.2. Primary and secondary outcomes

Primary and secondary outcomes are presented in [Table 1](#). There was a suggestion that RCTs reporting some large ES in their abstract were less likely to have a prospectively registered protocol compared with RCTs reporting nonlarge ES (45% vs 61%, P value = .0054).

RCTs reporting some large ES in their abstract were less likely to have any registered protocol (prospectively or retrospectively) compared with RCTs reporting nonlarge ES (74% vs 87%, P value = .0028).

Considering either publicly available data or data placed in a repository, overall data availability in included trials were very low and not different (P = .66) between the two groups. Only 15 trials had readily available raw data (6 in large ES group and 9 in the nonlarge ES group) and additional 7 trials (3 and 4 in the two groups, respectively) specified that the raw data had been already posted in a declared repository (with the repository containing information on the respective trial).

In sensitivity analyses for the primary outcome, difference in prospectively registered protocol rates amounted to 49% in large ES trials vs 61% in nonlarge ES trials (P = .034) when preferring registry-listed information, and to 45% vs 58% (P = .020) when considering all trials with contradictory date information to be retrospectively registered.

3.3. Additional outcomes

Additional exploratory analyses of methodological features of included trials indicated ([Table 2](#)) that RCTs reporting large ES in their abstract appeared to differ from nonlarge ES RCTs in several aspects: SAP (P = .0216), multicenter-trial (P = .0042), high-income country of corresponding author (P = .0001), and high-income country

conduct site (P = .0003). Reported effect sizes in large ES trials were significantly less likely to result from between-group comparisons that pertained to the randomized comparison (P < .0001) and tended to report less frequently allocation concealment (P = .0351), and significantly more likely resulted from nondrug/nonpsychological interventions (P = .0001). Furthermore, RCTs reporting large ES in their abstract tended to have lower sample size (median 60 (IQR: 32–110) vs 137 (IQR: 71–262), P < .0001 by Mann–Whitney U test).

Conversely, journal impact factor did not significantly differ between groups (median 3.8 (IQR: 2.6–4.8) in large vs (4.2 (IQR: 2.8–6)) in nonlarge ES trials, P = .101 by Mann–Whitney U test). Blinding and objective outcome did not differ between groups and early stopping due to effectiveness in interim analysis was equally rare. Also, features of data sharing were similar in both compared groups of trials and data were uncommonly shared ([Table 2](#)).

3.4. Subgroup analyses

Subgroup analyses appear in [Supplementary Table 3](#). Results for the primary outcomes were generally not different beyond chance between subgroups, although the difference between large ES and nonlarge ES trials for any protocol registration was more pronounced in studies with drug or psychological interventions (P = .022) than in studies with other interventions.

3.5. Regression analyses

In multivariable logistic regression, only between-group comparison ES was identified as a significant predictor of prospective protocol registration. Studies reporting ES reflecting the comparison between the randomized arms were more likely to have preregistered protocols (OR = 1.96, 95% CI: 1.11–3.54, P = .022) (see [Supplementary Table 4](#) for all regression analyses results).

3.6. Alternative effect size thresholds (0.5, 1.0, and 1.5) for primary outcomes

Prospective registration had suggestive differences between large ES and nonlarge ES trials even with alternative

Table 1. Primary and secondary outcomes

Outcomes (n, %)	Large ES trials N = 152	Nonlarge ES trials N = 175	RR (95%-CI)	P value
Primary outcome				
Prospective protocol registration	69 (45.4%)	107 (61.1%)	0.74 (0.60–0.92)	0.0054
Secondary outcomes				
Any protocol registration	113 (74.3%)	153 (87.4%)	0.85 (0.76–0.95)	0.0028
Raw data available or placed in repository	9 (5.9%)	13 (7.4%)	0.80 (0.35–1.81)	0.66

CI, confidence interval; ES, effect size; RR, risk ratio.

Table 2. Additional outcomes

Features, n (%)	Large ES trials	Nonlarge ES trials	RR (95%-CI)	P value
SAP	6 (3.9%)	19 (10.9%)	0.36 (0.15-0.89)	0.022
Multicenter	18 (11.8%)	43 (24.6%)	0.48 (0.29-0.80)	0.0042
High-income country author	94 (61.8%)	142 (81.1%)	0.76 (0.66-0.88)	0.0001
High-income country location	89 (58.6%)	135 (77.1%)	0.76 (0.65-0.89)	0.0003
Between-group ES	104 (68.4%)	156 (89.1%)	0.77 (0.68-0.86)	<0.0001
Abstract statement on s.s.	107 (70.4%)	117 (66.9%)	1.05 (0.91-1.22)	0.55
Intervention—Psychological	78 (51.3%)	110 (62.9%)	0.82 (0.67-0.99)	0.043
Intervention—Drug	20 (13.2%)	37 (21.1%)	0.62 (0.38-1.02)	0.079
Intervention—Other	54 (35.5%)	28 (16%)	2.22 (1.49-3.32)	0.0001
Corresponding author has any retracted article(s)—yes	6 (3.9%)	6 (3.4%)	1.15 (0.38-3.49)	1.0000
Blinding	93 (61.2%)	94 (53.7%)	1.14 (0.95-1.37)	0.181
Allocation concealment	65 (42.8%)	96 (54.9%)	0.78 (0.62-0.98)	0.035
Objective outcome	46 (30.3%)	40 (22.9%)	1.32 (0.92-1.91)	0.133
Early stopping due to effectiveness in interim analysis	1 (0.7%)	0 (0%)	Inf (NA)	0.465
Data sharing features				
Any data sharing statement	100 (65.8%)	125 (71.4%)	0.92 (0.79-1.07)	0.28
Access-request to authors	74 (48.7%)	90 (51.4%)	0.95 (0.76-1.18)	0.66
Explicit statement that data will not be shared	8 (5.3%)	9 (5.1%)	1.02 (0.40-2.59)	1
Access-data available in public now	6 (100%)	9 (90%)	1.11 (0.90-1.37)	1
Repository statement—intention	10 (6.6%)	20 (11.4%)	0.58 (0.28-1.19)	0.18
Repository placement: Repository contains information about the data/study	7 (77.8%)	12 (100%)	0.78 (0.55-1.10)	0.17
Repository access: Data are accessible on repository now	5 (71.4%)	8 (100%)	0.71 (0.45-1.14)	0.20

ES, effect size; RR, risk ratio; SAP, statistical analysis plan; s.s., statistical significance.

effect size thresholds: 50% vs 68% ($P = .010$), 45% vs 58% ($P = .025$), 43% vs 56% ($P = .10$) for cutoffs 0.5, 1.0, and 1.5, respectively. Respective results for rates of any protocol were 78% vs 91% ($P = .015$), 73% vs 85% ($P = .015$), and 75% vs 82% ($P = .25$). Data sharing was low in all groups regardless of cutoff chosen.

4. Discussion

RCTs presenting large ESs in their abstracts had suggestively lower rates of prospective registration and significantly lower rates of any protocol availability compared with studies where all the presented ESs in their abstract were not large in magnitude. Several other features were also associated with reporting large ES, most prominently smaller sample size, lesser likelihood of having multicenter design or SAP, reporting allocation concealment, estimating ES from comparisons not protected by randomization, and lesser chance that the corresponding

author is from a non-high-income country and/or the trial was done in a non-high-income country. All of these features may be indirect indicators that may herald less trustworthiness and/or transparency and are partially correlated among themselves. Finally, although declarations of intent to share data were common, real public data availability or repository placement were rare for all groups of assessed RCTs.

A previous meta-epidemiological study [20] has also found that across 32 meta-analyses with a total of 165 trials, prospectively registered trials had on average 19% smaller ESs than unregistered ones (relative odds ratio 0.81, 95% CI, 0.65–1.02), while across 37 meta-analyses with a total of 177 trials, those registered at any point had on average 15% smaller ESs than those never registered (relative odds ratio, 0.85, 95% CI, 0.67–1.08). However, these differences were not beyond chance even at traditional $P < .05$ thresholds, and most trials in this assessment were fairly old. A smaller meta-epidemiological evaluation of 16 meta-analyses (83 trials)

in orthodontics found significantly smaller effects in registered vs unregistered trials (0.36 points in SMD scale) [21]. These findings may reflect that registration may work toward reducing some biases, for example, those related to selective reporting. Alternatively, registration may be a quality marker that coexists with other design, conduct, and reporting practices that may decrease biases. Specific causal claims should be cautious. Some other empirical evaluations have noted that nonregistered trials may find more often significant, “positive” results and conclusions [22], but this was not an ubiquitous finding, for example, no such association between registration and positive conclusions has been seen in oncology trials [23] or trials of diverse topics [24].

Previous evaluations used older trials than our evaluation, and registration was less common in the past [25]. Despite increases in the registration rate over time, and also in prospective registration [25,26], many trials are still not prospectively registered. Based on the inconsistencies we noted in manuscripts vs registries, we cannot exclude those additional trials that seem to be prospectively registered may not be so. Lack of proper registration may allow many opportunities for selective reporting and other biases and may facilitate the reporting of large ESs, some of which may not be credible. Previous work has found that registration (and even more so prospective registration) may be associated with lower risk of bias in features like randomization sequence and blinding [27]. Registration well after a study has started or has even been completed remains common [28], but this may not afford similar protection from bias.

Furthermore, even a prospectively registered trial may have many analytical and reporting degrees of freedom, as the registrations may lack sufficient detail. More detail and better rigor may be afforded by the early posting/publication of a full SAP [29–34]. Large ES trials seemed to fail more prominently also on this front. Smaller sample size, single-center recruitment and provenance from non–high-income countries may also increase the chances that bias may be introduced. Disregarding the randomized between-group comparison and presenting within-group effects is a common way to inflate effect sizes. This practice correlated with studies lacking preregistration or any registration and warrants particular scrutiny.

There is no perfect indicator of trustworthiness for RCTs and all these markers discussed here may be seen as surrogates for protection or susceptibility to diverse biases. Availability of raw data, conversely, may be seen as a more direct measure of transparency. Raw data availability has increased from almost 0% in the beginning of the century to approximately 20% in biomedicine [13]. Progress has been made also for sharing of data from RCTs [35–39], but the rate remains low [38,40]. In our sample of trials, rate of sharing was very low for all trials, regardless of what ESs were reported in the abstract. In the absence of raw data transparency, some authors have raised wider

credibility concerns. For example, the experience of John Carlisle as editor of *Anesthesia* suggested that 30%–40% of trials submitted to this respectable journal have been “zombie trials”, that is, have implausible data that must be either wrong or fake [41]. Progress in data transparency may help alleviate these serious concerns in the future.

Our work has several limitations. First, protocol registration status may only represent an approximation to rigorous methodological process. Protocol detail and requirements of (national) trial registries vary substantially and implementation also varies across trials. Second, we did not compare methodological detail of protocols to conducted and reported methods in resulting trials. There is large potential for methodological deviations from protocol even with prospective registration [42]. Third, some studies failed to provide recruitment or study initiation dates within the text or provide conflicting dates within manuscript and registry-listed information, carrying a risk of misclassification. Sensitivity analyses generally supported our main findings. Fourth, we focused on trials that use SMD, and our findings may not generalize to other trials, for example, those with binary metrics. Fifth, reporting of effect sizes in general—and in abstracts in particular—may be more common in specific (sub-) areas of medical research. However, if anything, abstracts that do not report any ESs may be even less rigorous.

Allowing for these caveats, we conclude that RCTs presenting large ESs in their abstracts are more likely to lack features of transparency and trustworthiness. Large ESs in RCTs may superficially seem to be a sign of “success”, but in fact extreme results should be regarded as a sign of possible methodological shortcomings and deserve increased attention and careful scrutiny.

CRedit authorship contribution statement

Jonathan F. Henssler: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joana Reis-Pardal:** Writing – review & editing, Investigation, Data curation. **Lina Koppel:** Writing – review & editing, Investigation, Data curation. **John P.A. Ioannidis:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

There are no conflicts of interest to declare.

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Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112215>.

Data availability

The data included in the analyses are available from the authors upon request.

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